Today: Vector Field Following

Approaches:

1) Velocity Control
2) Point Control
3) Non-linear Control

I Define desired velocities

- Given a robot and desired vector to travel
  - let g represent velocity vector

A) Idea:

1) Adjust steering angle to align
2) Adjust our tangential velocity to match vector magnitude

$\underline{V_d = \|g\|}$

We will start looking at orientation error

$\theta_d = \tan^{-1}\left(\frac{g_2}{g_1}\right)$ ← can cause issues

$\underline{\omega_d = -K\theta_e}$

$\Rightarrow \underline{\theta_d = \operatorname{atan2}(g_2, g_1)}$

$\underline{\theta_e = \theta - \theta_d}$ → Does not garantee shortest rotation because θ is not unique

To solve this issue

We want θ to be in $[-\pi\ \pi]$

1) Form a vector θ_e
2) Use atan2 to get $\theta_e \in [-\pi, \pi]$

$\operatorname{atan2}(s\theta_e, c\theta_e)$

Works well w/ unicycle and differential

$\omega = \frac{V}{L}\tan(\phi)$ for $V \neq 0$

II Point Control

We don't care about the position, just velocity vector of the point.

$\dot{q}_1 = g \qquad \dot{g} = 0$

1) Design transformation $\dot{x} = q_c \quad x_d = g \quad z = \dot{q}_c - g$
2) Create control law
3) Use algebraic relationship to convert u_c to $\begin{bmatrix} a \\ \alpha \end{bmatrix}$

III Non-linear Control

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} V\cos\theta \\ V\sin\theta \\ \omega \end{bmatrix} \qquad h = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}$$

$$\dot{h} = \begin{bmatrix} -\omega s\theta \\ \omega c\theta \end{bmatrix} = \omega J h$$

$$p = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$p = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = vh$$

$$\Rightarrow \quad v = -k_v g^T h \qquad \omega = -k_\omega g^T J h$$